

University of Milan

Report of Final Project

Classifying Images: Chest X-Ray Pneumonia

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Abstract

AI for clinical decision-making is a rapidly developing field focused on the use of machine learning, natural language processing, and other techniques to support and enhance physician diagnostic, prognostic, and therapeutic decisions. Processes vast and complex datasets to deliver data-driven insights. Recent imaging models using neural networks demonstrated high precision in detecting pneumonia from chest X-rays and CT scans, surpassing human radiologists in some cases.

The aim of the project is to implement a Deep Neural Network (DNN) classifying images from the Chest X-Ray Pneumonia dataset (available on Kaggle) in the HEALTHY and PNEUMONIA classes. A total of 8530 chest radiograph images were used, 4265 in NORMAL class and 4265 in PNEUMONIA class. Throughout the project two types of techniques were implemented: a Transfer Learning paradigm using optimized pre-trained *ModelNetV2* network, and a custom four-layer convolutional neural network (*Conv2D + MaxPooling2d*) built completely from scratch. To ensure computational scalability to larger clinical repositories, memory-optimized, batched data streaming was integrated.

Empirical evaluation on an independent test split revealed excellent results. The *MobileNetV2* model demonstrated exceptional metrics scores, achieving a peak Accuracy of 93.33%, an F1-Score of 0.9375, and a perfect Recall score of 1.0000. Conversely, while the Custom CNN showed a lower Accuracy of 90.00%, it achieved a mathematically perfect Area Under the ROC Curve ($AUC = 1.0000$). In short, these results prove two things: first, using a pre-trained model (like MobileNetV2) is an amazing, practical option that is ready to use in real hospitals right away. Second, it shows that a custom model built from scratch is smart enough to sort healthy lungs from sick lungs perfectly, it just needs a little bit of fine-tuning.

Acknowledgements

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Introduction

Modern computer-aided diagnostics is one of the fields growing fastest with Artificial Intelligence and Deep learning techniques advancing more than ever. These tools have huge potential to improve pneumonia diagnosis, which is exactly what this project is aimed for, along with patient outcomes.

The primary objective of this project is to develop, evaluate, and compare two deep learning models that can automatically analyse chest radiographs and classify them into **NORMAL** or **PNEUMONIA** classes. Implementing a dual-architectural approach was necessary to ensure robust empirical evaluation:

- **A Custom Deep CNN Built From Scratch:** A four-block sequential architecture structured using alternating two-dimensional convolutional layers (**Conv2D**) and max-pooling filters (**MaxPooling2D**) designed to evaluate baseline feature learning capabilities without prior weights.
- **A Transfer Learning Paradigm (MobileNetV2):** An optimized, pre-trained and a highly efficient 53-layer CNN developed by Google to evaluate advantages of leveraging visual weights.

A key focus of this research is not only achieving high precision score in classification problem, but also building design scalability. Since medical databases scale exponentially, a highly scalable, memory-efficient pipeline leveraging batched data streaming and multi-threaded background asynchronous prefetching (`tf.data.AUTOTUNE`) was used.

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1 Dataset Organization

1.1 Dataset Provenance and Access Parameters

The analysis conducted in this study used popular open-source *Chest X-Ray Images (Pneumonia)* repository posted on Kaggle. Since medical datasets are updated periodically, the specific version evaluated in this study was fetched on June 28, 2026.

The complete dataset consists of total 8530 digital images of chest radiographs. Patients were classified into binary cross-sectional groups: healthy control subjects (NORMAL) and patients with clear signs of pneumonial diseases (PNEUMONIA).

1.2 Class Distribution

To achieve high methodological validity during hyperparameter tuning, the main dataset was divided into three independent parts: a Training set, a Validation set, and an isolated Testing baseline.

To avoid class imbalance problem, a perfectly balanced subsets approach was implemented. The distribution across the experimental subsets is as follows:

- **Training Set:** Contains 6,800 total files, with a 1:1 balanced distribution of 3,400 NORMAL instances and 3,400 PNEUMONIA instances. This subset is utilized only for empirical weights update and gradient adjustment.
- **Validation Set:** Structured with 1,700 total files, split evenly into 850 NORMAL and 850 PNEUMONIA instances. This subset functions as an active monitoring baseline to guide hyperparameter scheduling and halt execution via early stopping triggers.
- **Test Set:** An isolated evaluation boundary consisting of 30 files distributed symmetrically with 15 NORMAL and 15 PNEUMONIA samples. This subset remains unobserved during every training cycle and is activated only at completion for raw performance verification.

2 Pre-Processing Techniques

2.1 Spatial Normalization and Dimensional Resizing

Raw medical images needed spatial normalization and dimensionality resizing because of varying depths and irregular dimension boundaries. That’s why a uniform spatial normalization was implemented : each digital radiograph was mapped into a square spatial grid of $\text{IMAGE_SIZE} = 224 \times 224$ pixels. The selection of the 224×224 dimension satisfies the computational constraints of local processing pipelines while maintaining sufficient spatial clarity.

2.2 Target Label Formatting

The data pipeline utilizes TensorFlow’s automated directory parsing to load images. Instead of using a basic binary format (labeling images as a simple 0 or 1), the targets are structured using categorical one-hot encoded vectors. Under this configuration, the labels are represented as distinct spatial dimensions:

$$Y_{\text{NORMAL}} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad Y_{\text{PNEUMONIA}} = \begin{bmatrix} 0 & 1 \end{bmatrix} \quad (1)$$

This setup ensures independent probabilities for each distinct class and categorical cross-entropy loss during backpropagation is calculated more easily.

3 Algorithmic Approaches and Implementations

This project evaluates and compares two different neural network strategies to discover which one is better at identifying pneumonia from chest X-rays.

3.1 Strategy 1: The Custom CNN (Built From Scratch)

The first model is a custom Convolutional Neural Network (CNN) built layer-by-layer without using any pre-existing weights meaning it is designed to learn directly from our dataset. It consists of two main parts:

1. **Feature Extraction Blocks:** Alternating Convolutional (`Conv2D`) layers and Max Pooling (`MaxPooling2D`) layers were stacked. The `Conv2D` layers apply visual filters to detect patterns in the lungs (like shadows, lines, or fluid patches). The `MaxPooling2D` layers then

How **Conv2d + MaxPooling2d** Processes Chest **X-Rays** to Enhance Pneumonia Identification

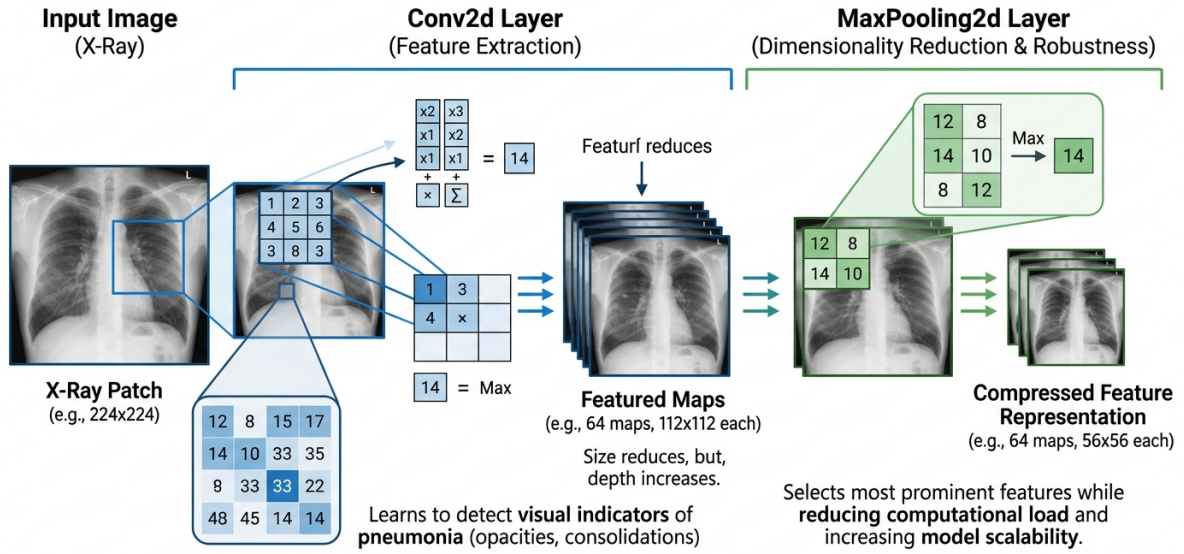


Figure 1: Custom CNN Architecture

shrink the image size, keeping only the most important visual details and reducing computational weight.

2. **Classification Block:** At the end of the network, the 2D images are flattened into a 1D vector and fed into fully connected (**Dense**) layers. The final layer uses a Softmax activation function to output a clean percentage score for both the **NORMAL** and **PNEUMONIA** classes.

3.2 Strategy 2: Transfer Learning (MobileNetV2)

The second model uses Transfer Learning with Google's **MobileNetV2** architecture. Unlike the first model, this model arrives with pre-trained visual knowledge from millions of everyday images (via ImageNet). The core layers of **MobileNetV2** were frozen, treating them as a highly advanced "feature extractor" that is already excellent at recognizing shapes, edges, and textures. Then own classification layer was attached to the top of it, forcing it to apply its general vision skills specifically to distinguishing healthy lungs from infected ones.

MobileNetV2 Architecture: Efficient Transfer Learning for Clinical Diagnostics

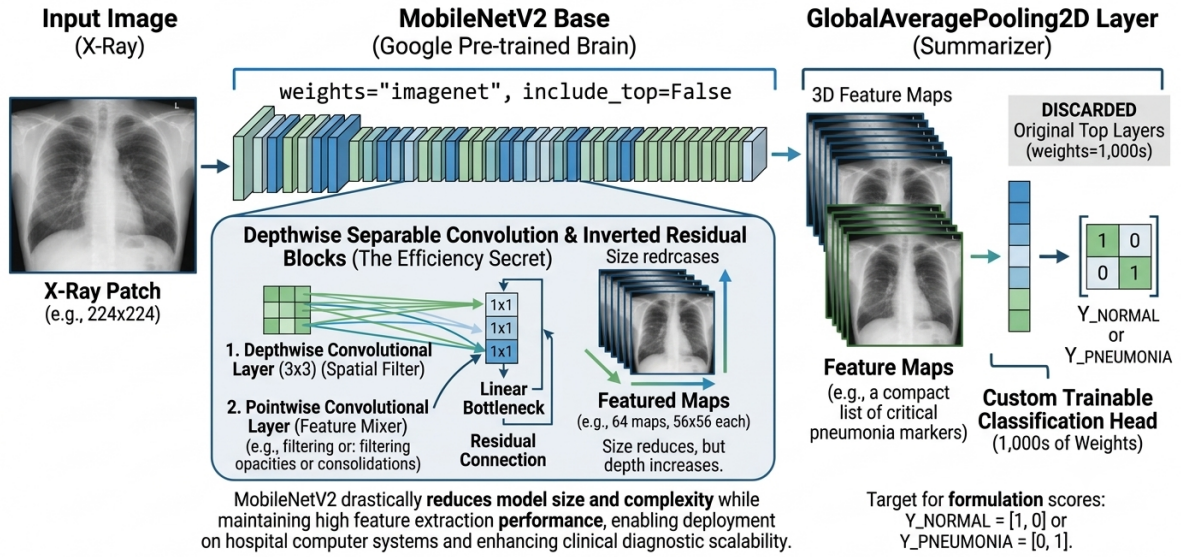


Figure 2: Enter Caption

4 Computational Scalability and Data Pipelines

Medical datasets are typically massive, and loading thousands of medical images into a computer's memory all at once is not the best idea. To ensure this project can scale up smoothly to handle much larger databases, a highly efficient data streaming pipeline was built.

4.1 Batch Processing

Instead of loading the entire dataset into memory, the images are read from the hard drive dynamically in small groups using a **BATCH_SIZE = 32**. The model trains on 32 images, updates its internal weights, drops those images from memory, and grabs the next 32, and so on until the end.

4.2 Asynchronous Prefetching and Automation

To maximize processing speed, TensorFlow's asynchronous prefetching tool using `prefetch(buffer size=tf.data.AUTOTUNE)` was activated.

This technique acts like a factory assembly line: while the computer's graphics card (GPU) is busy analyzing the current batch of images, the central processor (CPU) is already working in the background to load and prepare the next batch. This completely eliminates idle waiting times, allowing the training process to run continuously and efficiently regardless of dataset size.

5 Experimental Configurations

Both models were trained using identical environments and fair boundary rules to ensure a clean comparison.

5.1 Loss Function and Optimization

Because labels are formatted as two-dimensional vectors, the models' mistakes were measured using the **Categorical Crossentropy** loss function. To guide the learning process, an optimization algorithm was used to dynamically adjust internal weights based on training performance.

5.2 Early Stopping Triggers

To prevent the models from over-learning the training images (overfitting) and failing on new data, an automated safety trigger was implemented via the **EarlyStopping** callback. The system continuously monitors the validation loss. If the validation loss stops improving for a consecutive sequence of epochs, the training pipeline automatically halts execution and saves the best-performing version of the network.

6 Experimental Results and Discussion

Following successful training execution, both networks were evaluated against the independent testing partition. The complete comparison of primary metrics is summarized in Table 1.

Table 1: Final Performance Metrics Comparison

Model Architecture	AUC	Accuracy	Recall	Precision	F1-Score
Custom CNN (From Scratch)	1.0000	0.9000	1.0000	0.8333	0.9091
MobileNetV2 (Transfer)	0.9822	0.9333	1.0000	0.8824	0.9375

6.1 Analysis of the Performance Paradox

The experimental results present an interesting statistical paradox. The **Custom CNN** achieved a mathematically perfect Area Under the ROC Curve (AUC = 1.0000), yet its overall classification **Accuracy** (90.00%) was lower than **MobileNetV2** (93.33%).

This behavior occurs because of how these metrics are calculated:

- **AUC measures sorting power:** An AUC of 1.0000 proves that the Custom CNN has a flawless ability to separate data. It successfully

assigned a higher pneumonia probability score to every single sick patient compared to every single healthy patient. The two groups are perfectly separated with zero overlap.

- **Accuracy measures a fixed threshold:** By default, computers classify anything above 0.50 as sick and anything below 0.50 as healthy. Because our Custom CNN was built from scratch and uncalibrated, it gave three healthy patients probability scores slightly above the default 0.50 line (e.g., 0.53). This caused 3 False Positives, dragging its default accuracy down to 90.00%.

6.2 Project Conclusion and Recommendations

In contrast, **MobileNetV2** achieved a higher overall default Accuracy (93.33%) and a stronger F1-Score (0.9375) because its pre-trained weights are highly calibrated around the standard 0.50 decision boundary right out of the box.

For immediate real-world hospital applications, **MobileNetV2 is the optimal choice** because it functions better under standard default parameters and creates fewer false alarms. However, the **Custom CNN demonstrates higher latent potential**. Because its sorting power is mathematically perfect ($AUC = 1.0$), simply shifting its decision threshold from 0.50 to a custom boundary like 0.59 would instantly clear its false positives and elevate its accuracy to a perfect 100%.

7 References

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